MILITARY SPECIFICATION

RESISTORS, VARIABLE, COMPOSITION, STYLE RV4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RV4 composition, variable, resistors.

2. APPLICABLE DOCUMENT

2.1 Government document.

2.1.1 Specification. Unless otherwise specified, the following specification of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-94 - Resistors, Variable, Composition, General Specification for.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-94, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Mounting and locking nuts. The mounting nut shall be 3/32-inch thick and unless otherwise specified, shall measure 1/2 inch across the hexagonal flats. The locking nuts shall be within the limits of 0.151 and 0.234-inch thick and shall measure 1/2-inch across the hexagonal flats. Thread size shall be 3/8-32NEF-2B.

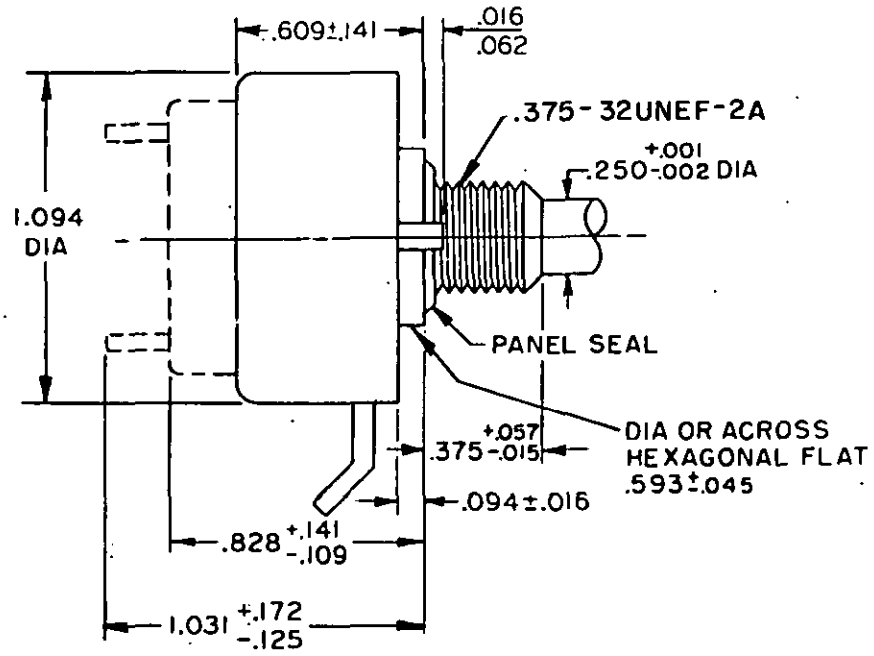
3.2.2 Internal-tooth lockwasher. An internal-tooth lockwasher of suitable size shall be supplied.

3.2.3 Retainer rings. If retainer rings are used, they shall be not thicker than 0.047 inch.

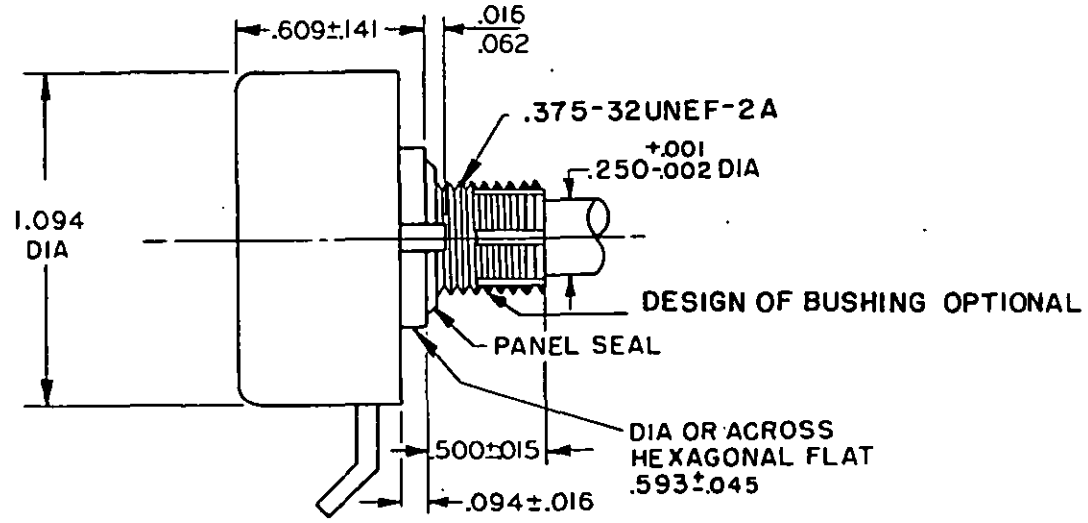
3.3 Total mechanical rotation. The total mechanical rotation with switch shall be within the limits of 330° and 340°; without switch, within the limits of 309° and 320°.

3.4 Shaft length. The length of operating shaft shall be in accordance with table I.

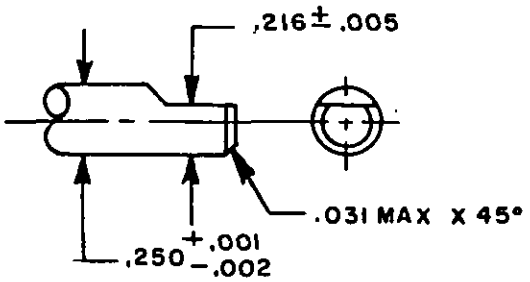
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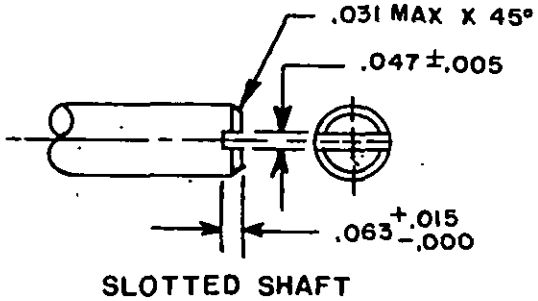
SHAFT- AND PANEL-SEALED-STANDARD BUSHING TYPE



SHAFT- AND PANEL-SEALED-LOCKING BUSHING TYPE



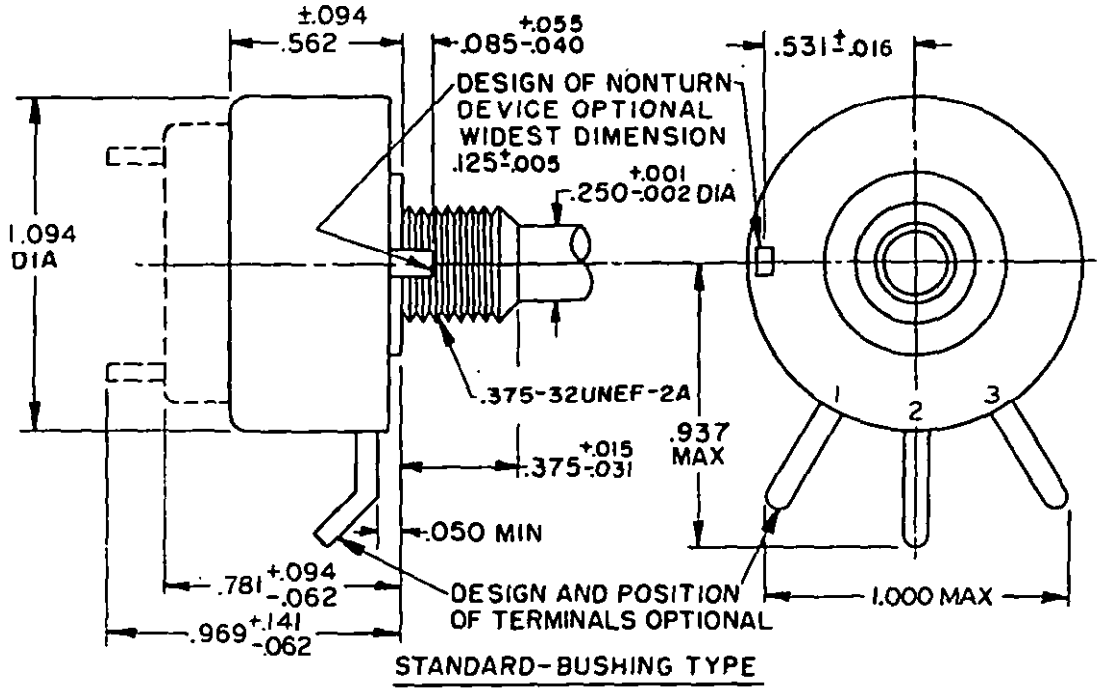
FLATTED SHAFT



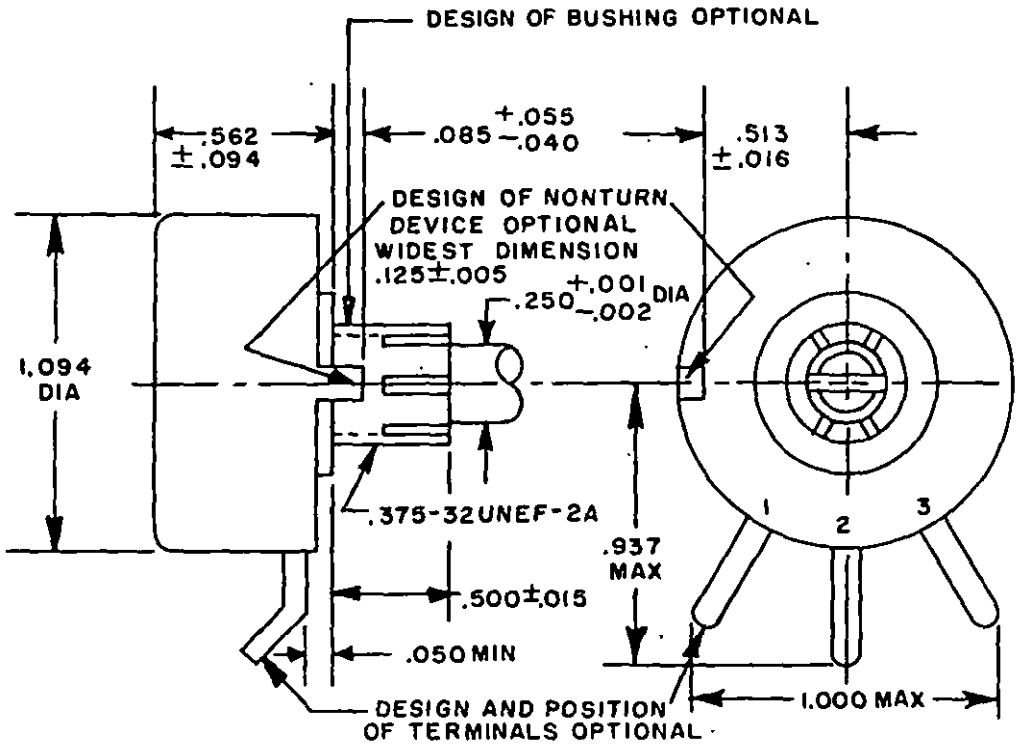
SLOTTED SHAFT

FIGURE 1. Style RV4 resistors.

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STANDARD-BUSHING TYPE



LOCKING-BUSHING TYPE

FIGURE 1. Style RV4 resistors- Continued.

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INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.057	1.45	.216	5.49
.002	.05	.062	1.57	.250	6.35
.005	.13	.063	1.60	.375	9.53
.015	.38	.094	2.39	.500	12.70
.016	.41	.109	2.77	.593	15.06
.031	.79	.125	3.18	.609	15.47
.045	1.14	.141	3.58	.828	21.03
.047	1.19	.172	4.37	1.031	26.19
				1.094	27.79

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4mm.
3. Unless otherwise specified, tolerance is +.062(1.57mm).

FIGURE 1. Style RV4 resistors- Continued.

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* TABLE I. Length of operating shaft. 1/

Symbol	Length of operating shaft from mounting surface of resistor (nominal)		
	Flatted	Slotted	
	Bushings N and S (inches, ±1/32)	Bushings N and S (inches, ±1/32)	Bushings L and T (inches, ±1/32)
A	---	---	5/8
B	---	1/2	---
D	7/8	7/8	7/8 2/
E	---	---	---
G	---	1 1/4 2/	---
N	---	1 1/2 2/	---
J	---	2 2/	---
K	---	2 1/2 2/	---

1/ For resistors with panel seals, the mounting surface shall be considered the metal face of the panel seal when the seal is firmly seated against the resistor body.

2/ For replacement purposes only. Not for new design.

3.5 Nominal total resistance and rated continuous working voltage. The nominal total resistance and rated continuous working voltage (RCWV) shall be as specified in table II.

TABLE II. Nominal total resistance and rated continuous working voltage.

Nominal total resistance	RCWV 1/ (volts)		Nominal total resistance	RCWV 1/ (volts)		Nominal total resistance	RCWV 1/ (volts)	
	Taper A	Tapers C and F		Taper A	Tapers C and F		Taper A	Tapers C and F
Ohms			Ohms			Megohms		
50	10	---	5,000	100	71	.10	445	316
100	14	10	7,500	122	87	.15	500	350
150	17	12	10,000	141	100	.20 2/	500	350
200 2/	20	---	15,000	173	123	.25	500	350
250	22	16	20,000 2/	200	141	.35	500	350
350	26	18	25,000	224	160	.50	500	350
500	32	23	35,000	264	187	.75	500	350
750	39	28	50,000	316	224	1.0	500	350
1,000	45	32	75,000	387	249	1.5	500	350
1,500 2/	55	39				2.0	500	350
2,000	63.3	45				2.5	500	350
2,500	71	50				3.5	500	350
3,500	84	60				5.0	500	350

1/ Rated continuous working voltage at 70°C.

2/ For replacement purposes only. Not for new design.

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3.6 Power rating. The power rating shall be 2 watts for taper-A resistors and 1 watt for tapers-C and -F resistors.

3.7 Torque.

3.7.1 Operating. The torque required to effect rotation shall be not less than 1 ounce-inch nor more than 6 ounce-inches.

3.7.2 Stop. The torque applied through the operating shaft to the stops shall be 8 pound-inches.

3.7.3 Locking bushing. For the locking-bushing type, the locking nut shall be tightened with a torque of 10 pound-inches. After the locking nut is tightened, the contact arm shall not move when a torque of 30 ounce-inches is applied to the shaft.

3.9 Thermal cycling. The resistors shall not change in resistance in excess of 4 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-94, and as specified herein.

4.2 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential at reduced barometric pressure shall be 450 volts, rms.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-94.

6. NOTES

6.1 Notes. The notes specified in MIL-R-94 are applicable to this specification.

6.2 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - MI, AR
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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